

Chapter 15

System Timer

15.1 Overview

ESP32-P4 provides a 52-bit system timer, which can be used to generate tick interrupts for the operating system, or be used as a general timer to generate periodic interrupts or one-time interrupts.

15.2 Features

The system timer has the following features:

- Two 52-bit counters and three 52-bit comparators
- Software accessing registers clocked by APB_CLK
- CNT_CLK used for counting, with an average frequency of 16 MHz in two counting cycles
- 40 MHz XTAL_CLK as the clock source of CNT_CLK
- 52-bit alarm values (t) and 26-bit alarm periods (δt)
- Two modes to generate alarms:
 - Target mode: only a one-time alarm is generated based on the alarm value (t)
 - Period mode: periodic alarms are generated based on the alarm period (δt)
- Three comparators generating three independent interrupts based on configured alarm value (t) or alarm period (δt)
- Software configuring the reference count value. For example, the system timer is able to load back the sleep time recorded by RTC timer via software after Light-sleep
- Able to stall or continue running when CPU stalls or enters the on-chip-debugging mode
- Alarm for Event Task Matrix (ETM) event

15.3 System Timer Structure

The timer consists of two counters: UNIT0 and UNIT1. The count values can be monitored by three comparators, COMP0, COMP1, and COMP2. See the timer block diagram in Figure 15.3-1.

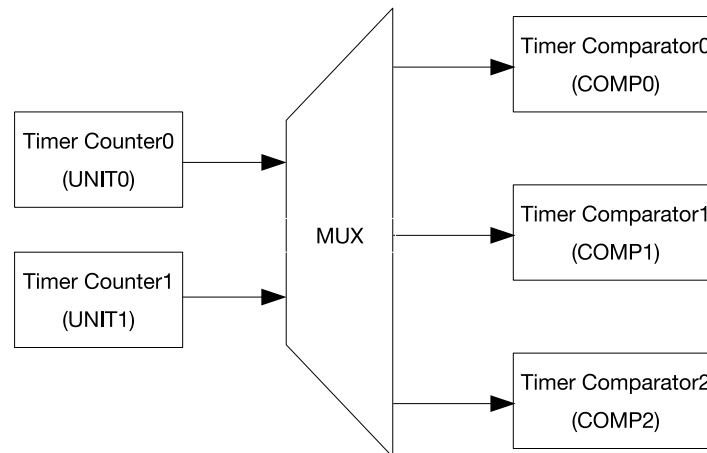


Figure 15.3-1. System Timer Structure

15.4 Clock Source Selection

The counters and comparators use XTAL_CLK or RC_FAST_CLK as the clock sources. The clock source can be selected by configuring field HP_SYS_CLKRST_REG_SYSTIMER_CLK_SRC_SEL in register [HP_SYS_CLKRST_PERI_CLK_CTRL21_REG](#). After XTAL_CLK is scaled by 2.5, a counter clock signal CNT_CLK is generated with a frequency of $f_{XTAL_CLK}/2.5$. The average clock frequency of CNT_CLK is 16 MHz, as shown in Figure 15.5-1. The timer counter is incremented by $1/16\ \mu s$ on each CNT_CLK cycle.

Software operation such as configuring registers is clocked by APB_CLK. For more information about APB_CLK, see Chapter 10 [Reset and Clock](#).

The following two bits of system registers are also used to control the system timer:

- Set HP_SYS_CLKRST_REG_SYSTIMER_APB_CLK_EN in register [HP_SYS_CLKRST_SOC_CLK_CTRL2_REG](#) to enable APB_CLK signal to the system timer.
- Set HP_SYS_CLKRST_REG_RST_EN_SYSTIMER in register [HP_SYS_CLKRST_HP_RST_EN1_REG](#) to reset the system timer.

Note that if the timer is reset, its registers will be restored to their default values. For more information, please refer to Chapter 10 [Reset and Clock](#).

15.5 Functional Description

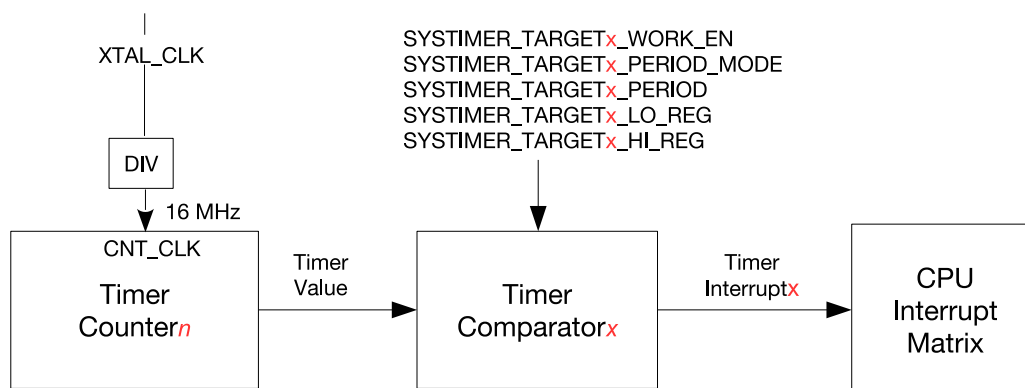


Figure 15.5-1. System Timer Alarm Generation

Figure 15.5-1 shows the procedure to generate alarm in system timer. In this process, one timer counter and one timer comparator are used. An alarm interrupt will be generated accordingly based on the comparison result in the comparator.

15.5.1 Counter

The system timer has two 52-bit timer counters, shown as UNIT n ($n = 0$ or 1). Their counting clock source is a 16 MHz clock, i.e., CNT_CLK. Whether UNIT n works or not is controlled by three bits in register SYSTIMER_CONF_REG:

- SYSTIMER_TIMER_UNIT n _WORK_EN: set this bit to enable the counter UNIT n in the system timer.
- SYSTIMER_TIMER_UNIT n _CORE0_STALL_EN: if this bit is set, the counter UNIT n stops when CPU0 is stalled. The counter continues its counting after the CPU resumes.
- SYSTIMER_TIMER_UNIT n _CORE1_STALL_EN: if this bit is set, the counter UNIT n stops when CPU1 is stalled. The counter continues its counting after the CPU resumes.

The configuration of the bits to control the counter UNIT n is shown below, assuming that CPU is stalled.

Table 15.5-1. UNIT n Configuration Bits

SYSTIMER_TIMER_UNIT n _WORK_EN	SYSTIMER_TIMER_UNIT n _CORE0_STALL_EN	SYSTIMER_TIMER_UNIT n _CORE1_STALL_EN	Counter UNIT n
0	x [*]	x [*]	Not at work
1	1	x	Stop counting, but will continue its counting after the CPU0 resumes
1	x	1	Stop counting, but will continue its counting after the CPU1 resumes
1	0	0	Keep counting

^{*} x: Don't-care.

When the counter UNIT n is at work, the count value is incremented on each counting cycle. When the counter UNIT n is stopped, the count value stops increasing and keeps unchanged.

The lower 32 and higher 20 bits of the initial count value are loaded from the registers SYSTIMER_TIMER_UNIT n _LOAD_LO and SYSTIMER_TIMER_UNIT n _LOAD_HI. Writing 1 to the bit SYSTIMER_TIMER_UNIT n _LOAD will trigger a reload event, and the current count value will be changed immediately. If UNIT n is at work, the counter will continue to count up from the new reloaded value.

Writing 1 to SYSTIMER_TIMER_UNIT n _UPDATE will trigger an update event. The lower 32 and higher 20 bits of the current count value will be locked into the registers SYSTIMER_TIMER_UNIT n _VALUE_LO and SYSTIMER_TIMER_UNIT n _VALUE_HI, and then SYSTIMER_TIMER_UNIT n _VALUE_VALID is asserted. Before the next update event, the values of SYSTIMER_TIMER_UNIT n _VALUE_LO and SYSTIMER_TIMER_UNIT n _VALUE_HI remain unchanged.

15.5.2 Comparator and Alarm

The system timer has three 52-bit comparators, shown as COMP x ($x = 0, 1, \text{ or } 2$). The comparators can generate independent interrupts based on different alarm values (t) or alarm periods (δt).

Configure SYSTIMER_TARGET x _PERIOD_MODE to choose from the two alarm modes for each COMP x :

- 1: period mode
- 0: target mode

In period mode, the alarm period (δt) is provided by the register SYSTIMER_TARGET x _PERIOD. Assuming that current count value is t_1 , when it reaches $(t_1 + \delta t)$, an alarm interrupt will be generated. When the count value reaches $(t_1 + 2 * \delta t)$, another alarm interrupt also will be generated. By such way, periodic alarms are generated.

In target mode, the lower 32 bits and higher 20 bits of the alarm value (t) are provided by SYSTIMER_TIMER_TARGET x _LO and SYSTIMER_TIMER_TARGET x _HI. Assuming that current count value is t_2 ($t_2 \leq t$), an alarm interrupt will be generated when the count value reaches the alarm value (t). Unlike in period mode, only one alarm interrupt is generated in target mode.

SYSTIMER_TARGET x _TIMER_UNIT_SEL is used to choose the count value from which timer counter to be compared to generate alarms:

- 1: Use the count value from UNIT1
- 0: Use the count value from UNIT0

Finally, set SYSTIMER_TARGET x _WORK_EN and COMP x starts to compare the count value:

- In target mode, COMP x compares it with the alarm value (t).
- In period mode, COMP x compares it with the alarm period ($t_1 + n * \delta t$).

An alarm is generated when the count value equals to the alarm value (t) in target mode or to the start value + $n * \text{alarm period } \delta t$ ($n = 1, 2, 3 \dots$) in period mode. But if the alarm value (t) set in registers is less than the current count value, i.e., the target has already passed, when the current count value is larger than the alarm value (t) within a range ($0 \sim 2^{51} - 1$), an alarm interrupt will also be generated immediately. No matter in target mode or period mode, the low 32 bits and high 20 bits of the real alarm value can always be read from SYSTIMER_TARGET x _LO_RO and SYSTIMER_TARGET x _HI_RO. The alarm trigger point and the relationship between current count value t_c and the alarm value t_t are shown below.

Table 15.5-2. Trigger Point

Relationship Between t_c and t_t	Trigger Point
$t_c - t_t \leq 0$	$t_c = t_t$, an alarm is triggered.
$0 \leq t_c - t_t < 2^{51} - 1$ ($t_c < 2^{51}$ and $t_t < 2^{51}$, or $t_c \geq 2^{51}$ and $t_t \geq 2^{51}$)	An alarm is triggered immediately.
$t_c - t_t \geq 2^{51} - 1$	t_c overflows after counting to its maximum value 52'hffffffffffff, and then starts counting up from 0. When its value reaches t_t , an alarm is triggered.

15.5.3 Event Task Matrix

The system timer on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows the system timer's ETM tasks to be triggered by any peripherals' ETM events, or system timer's ETM events to trigger any peripherals' ETM tasks. This section introduces the ETM tasks and events related to the system timer. For more information, please refer to Chapter 13 [Event Task Matrix \(ETM\)](#).

The system timer can generate the following ETM event:

- SYSTIMER_EVT_CNT_CMP x : Indicates the alarm events generated by COMP x .

When [SYSTIMER_ETM_EN](#) is set to 1, the alarm events can trigger the ETM event.

15.5.4 Synchronization Operation

The frequency of the clock used for software configuration is different from that of the clock driving the timer counters and comparators. Therefore, synchronization is required for configuring some registers. The steps are as follows:

1. Software writes specific values to configuration fields. See the first column in Table 15.5-3.
2. Software writes 1 to corresponding bits to start synchronization. See the second column in Table 15.5-3.

Table 15.5-3. Synchronization Operation for Configuration Registers

Configuration Fields	Synchronization Enable Bit
SYSTIMER_TIMER_UNIT n _LOAD_LO SYSTIMER_TIMER_UNIT n _LOAD_HI	SYSTIMER_TIMER_UNIT n _LOAD
SYSTIMER_TARGET x _PERIOD SYSTIMER_TIMER_TARGET x _HI SYSTIMER_TIMER_TARGET x _LO	SYSTIMER_TIMER_COMP x _LOAD

The frequency of the clock used for software reading is different from that of the clock driving the timer counters and comparators. Therefore, synchronization is also needed for reading some registers. The steps are as follows:

1. Software writes specific values to the updating register [SYSTIMER_TIMER_UNIT \$n\$ _UPDATE](#).

2. Software reads the corresponding bit `SYSTIMER_TIMER_UNIT $n$ _VALUE_VALID` to be valid to confirm synchronization is done.
3. Software reads the corresponding status registers `SYSTIMER_TIMER_UNIT $n$ _VALUE_HI` and `SYSTIMER_TIMER_UNIT $n$ _VALUE_LO`.

15.6 Interrupts

ESP32-P4's system timer can generate the following interrupt signals that will be sent to the [Interrupt Matrix](#).

- SYSTIMER_TARGET0_INT
- SYSTIMER_TARGET1_INT
- SYSTIMER_TARGET2_INT

There are several internal interrupt sources from the system timer that can generate the above interrupt signals. The interrupt sources from the system timer are listed with their trigger conditions and the resulted interrupt signals in Table 15.6-1.

Table 15.6-1. System Timer's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
SYSTIMER_INT0	When COMPO alarms	SYSTIMER_TARGET0_INT
SYSTIMER_INT1	When COMP1 alarms	SYSTIMER_TARGET1_INT
SYSTIMER_INT2	When COMP2 alarms	SYSTIMER_TARGET2_INT

The above interrupts are level-type alarm interrupts. The interrupt signal is asserted high when the comparator starts to alarm. Until the software clears the interrupt, it remains high. To enable interrupts, set the bit `SYSTIMER_TARGET $x$ _INT_ENA`.

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter 12 [Interrupt Matrix](#) > Section 12.2 [Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section 15.8 [Register Summary](#).

15.7 Programming Procedure

When configuring COMP x and UNIT n , please ensure the corresponding COMP and UNIT are at work.

15.7.1 Read Current Count Value

1. Set `SYSTIMER_TIMER_UNIT $n$ _UPDATE` to fill the current count value of COMP x into `SYSTIMER_TIMER_UNIT $n$ _VALUE_HI` and `SYSTIMER_TIMER_UNIT $n$ _VALUE_LO`.

2. Poll the reading of `SYSTIMER_TIMER_UNIT $n$ _VALUE_VALID` till it is 1. Then, user can read the count value from `SYSTIMER_TIMER_UNIT $n$ _VALUE_HI` and `SYSTIMER_TIMER_UNIT $n$ _VALUE_LO`.
3. Read the lower 32 bits and higher 20 bits from `SYSTIMER_TIMER_UNIT $n$ _VALUE_LO` and `SYSTIMER_TIMER_UNIT $n$ _VALUE_HI` respectively.

15.7.2 Configure a One-Time Alarm in Target Mode

1. Set `SYSTIMER_TARGET $x$ _TIMER_UNIT_SEL` to select the counter (UNIT 0 or UNIT 1) used for comparison with COMP x .
2. Read the current count value with reference to Section 15.7.1. This value will be used to calculate the alarm value (t) in Step 4.
3. Clear `SYSTIMER_TARGET $x$ _PERIOD_MODE` to enable target mode.
4. Set an alarm value (t), and fill its lower 32 bits into `SYSTIMER_TIMER_TARGET $x$ _LO`, and the higher 20 bits into `SYSTIMER_TIMER_TARGET $x$ _HI`.
5. Set `SYSTIMER_TIMER_COMP $x$ _LOAD` to synchronize the alarm value (t) to COMP x , i.e., load the alarm value (t) to the COMP x .
6. Set `SYSTIMER_TARGET $x$ _WORK_EN` to enable the selected COMP x . COMP x starts comparing the count value with the alarm value (t).
7. Set `SYSTIMER_TARGET $x$ _INT_ENA` to enable the timer interrupt. When UNIT n reaches the alarm value (t), a `SYSTIMER_TARGET $x$ _INT` interrupt is triggered.

15.7.3 Configure Periodic Alarms in Period Mode

1. Set `SYSTIMER_TARGET $x$ _TIMER_UNIT_SEL` to select the counter (UNIT 0 or UNIT 1) used for comparison with COMP x .
2. Set an alarm period (δt), and fill it into `SYSTIMER_TARGET $x$ _PERIOD`.
3. Set `SYSTIMER_TIMER_COMP $x$ _LOAD` to synchronize the alarm period (δt) to COMP x , i.e., load the alarm period (δt) to COMP x .
4. Clear and then set `SYSTIMER_TARGET $x$ _PERIOD_MODE` to configure COMP x into period mode.
5. Set `SYSTIMER_TARGET $x$ _WORK_EN` to enable the selected COMP x . COMP x starts comparing the count value with the sum of (start value + $n \cdot \delta t$) ($n = 1, 2, 3, \dots$).
6. Set `SYSTIMER_TARGET $x$ _INT_ENA` to enable the timer interrupt. A `SYSTIMER_TARGET $x$ _INT` interrupt is triggered when UNIT n reaches start value + $n \cdot \delta t$ ($n = 1, 2, 3, \dots$) set in Step 2.

15.7.4 Update After Light-sleep

1. Configure RTC timer before the chip goes to Light-sleep mode, to record the exact sleep time. For more information, see Chapter 14 *Low-Power Management*.
2. Read the sleep time from the RTC timer when the chip wakes up from Light-sleep mode.
3. Read the current count value of system timer, see Section 15.7.1.

4. Convert the time value recorded by RTC timer from the clock cycles based on RTC_SLOW_CLK to that based on 16 MHz CNT_CLK. For example, if the frequency of RTC_SLOW_CLK is 32 kHz, the recorded RTC timer value should be converted by multiplying by 500.
5. Add the converted RTC value to the current count value of system timer:
 - Fill the new value into SYSTIMER_TIMER_UNIT n _LOAD_LO (low 32 bits) and SYSTIMER_TIMER_UNIT n _LOAD_HI (high 20 bits).
 - Set SYSTIMER_TIMER_UNIT n _LOAD to load the new timer value into the system timer. By such way, the system timer is updated.

15.8 Register Summary

The addresses in this section are relative to system timer base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Clock Control Register			
SYSTIMER_CONF_REG	Configure system timer clock	0x0000	R/W
UNIT0 Control and Configuration Registers			
SYSTIMER_UNIT0_OP_REG	Read UNIT0 value to registers	0x0004	varies
SYSTIMER_UNIT0_LOAD_HI_REG	High 20 bits to be loaded to UNIT0	0x000C	R/W
SYSTIMER_UNIT0_LOAD_LO_REG	Low 32 bits to be loaded to UNIT0	0x0010	R/W
SYSTIMER_UNIT0_VALUE_HI_REG	UNIT0 value, high 20 bits	0x0040	RO
SYSTIMER_UNIT0_VALUE_LO_REG	UNIT0 value, low 32 bits	0x0044	RO
SYSTIMER_UNIT0_LOAD_REG	UNIT0 synchronization register	0x005C	WT
UNIT1 Control and Configuration Registers			
SYSTIMER_UNIT1_OP_REG	Read UNIT1 value to registers	0x0008	varies
SYSTIMER_UNIT1_LOAD_HI_REG	High 20 bits to be loaded to UNIT1	0x0014	R/W
SYSTIMER_UNIT1_LOAD_LO_REG	Low 32 bits to be loaded to UNIT1	0x0018	R/W
SYSTIMER_UNIT1_VALUE_HI_REG	UNIT1 value, high 20 bits	0x0048	RO
SYSTIMER_UNIT1_VALUE_LO_REG	UNIT1 value, low 32 bits	0x004C	RO
SYSTIMER_UNIT1_LOAD_REG	UNIT1 synchronization register	0x0060	WT
Comparator0 Control and Configuration Registers			
SYSTIMER_TARGET0_HI_REG	Alarm value to be loaded to COMPO, high 20 bits	0x001C	R/W
SYSTIMER_TARGET0_LO_REG	Alarm value to be loaded to COMPO, low 32 bits	0x0020	R/W
SYSTIMER_TARGET0_CONF_REG	Configure COMPO alarm mode	0x0034	R/W
SYSTIMER_COMPO_LOAD_REG	COMPO synchronization register	0x0050	WT
Comparator1 Control and Configuration Registers			
SYSTIMER_TARGET1_HI_REG	Alarm value to be loaded to COMP1, high 20 bits	0x0024	R/W
SYSTIMER_TARGET1_LO_REG	Alarm value to be loaded to COMP1, low 32 bits	0x0028	R/W
SYSTIMER_TARGET1_CONF_REG	Configure COMP1 alarm mode	0x0038	R/W
SYSTIMER_COMP1_LOAD_REG	COMP1 synchronization register	0x0054	WT
Comparator2 Control and Configuration Registers			
SYSTIMER_TARGET2_HI_REG	Alarm value to be loaded to COMP2, high 20 bits	0x002C	R/W
SYSTIMER_TARGET2_LO_REG	Alarm value to be loaded to COMP2, low 32 bits	0x0030	R/W
SYSTIMER_TARGET2_CONF_REG	Configure COMP2 alarm mode	0x003C	R/W
SYSTIMER_COMP2_LOAD_REG	COMP2 synchronization register	0x0058	WT
Interrupt Registers			
SYSTIMER_INT_ENA_REG	Interrupt enable register of system timer	0x0064	R/W
SYSTIMER_INT_RAW_REG	Interrupt raw register of system timer	0x0068	R/W/TC/SS
SYSTIMER_INT_CLR_REG	Interrupt clear register of system timer	0x006C	WT

Name	Description	Address	Access
SYSTIMER_INT_ST_REG	Interrupt status register of system timer	0x0070	RO
COMP0 Status Registers			
SYSTIMER_REAL_TARGET0_LO_REG	Actual target value of COMP0, low 32 bits	0x0074	RO
SYSTIMER_REAL_TARGET0_HI_REG	Actual target value of COMP0, high 20 bits	0x0078	RO
COMP1 Status Registers			
SYSTIMER_REAL_TARGET1_LO_REG	Actual target value of COMP1, low 32 bits	0x007C	RO
SYSTIMER_REAL_TARGET1_HI_REG	Actual target value of COMP1, high 20 bits	0x0080	RO
COMP2 Status Registers			
SYSTIMER_REAL_TARGET2_LO_REG	Actual target value of COMP2, low 32 bits	0x0084	RO
SYSTIMER_REAL_TARGET2_HI_REG	Actual target value of COMP2, high 20 bits	0x0088	RO
Version Register			
SYSTIMER_DATE_REG	Version control register	0x00FC	R/W

15.9 Registers

The addresses in this section are relative to system timer base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

Register 15.1. SYSTIMER_CONF_REG (0x0000)

[illegible]

SYSTIMER_ETM_EN Configures whether to enable generation of ETM events.

0: Disable

1: Enable

(R/W)

SYSTIMER_TARGET2_WORK_EN Configures whether to enable COMP2.

0: Disable

1: Enable

(R/W)

SYSTIMER_TARGET1_WORK_EN Configures whether to enable COMP1. See details in [SYSTIMER_TARGET2_WORK_EN](#). (R/W)

SYSTIMER_TARGET0_WORK_EN Configures whether to enable COMPO. See details in [SYSTIMER_TARGET2_WORK_EN](#). (R/W)

SYSTIMER_TIMER_UNIT1_CORE1_STALL_EN Configures whether UNIT1 is stalled when CORE1 is stalled.

0: UNIT1 is not stalled.

1: UNIT1 is stalled.

(R/W)

SYSTIMER_TIMER_UNIT1_CORE0_STALL_EN Configures whether UNIT1 is stalled when CORE0 is stalled. See details in [SYSTIMER_TIMER_UNIT1_CORE1_STALL_EN](#). (R/W)

SYSTIMER_TIMER_UNIT0_CORE1_STALL_EN Configures whether UNIT0 is stalled when CORE1 is stalled. See details in [SYSTIMER_TIMER_UNIT1_CORE1_STALL_EN](#). (R/W)

SYSTIMER_TIMER_UNIT0_CORE0_STALL_EN Configures whether UNIT0 is stalled when CORE0 is stalled. See details in [SYSTIMER_TIMER_UNIT1_CORE1_STALL_EN](#). (R/W)

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Register 15.1. SYSTIMER_CONF_REG (0x0000)

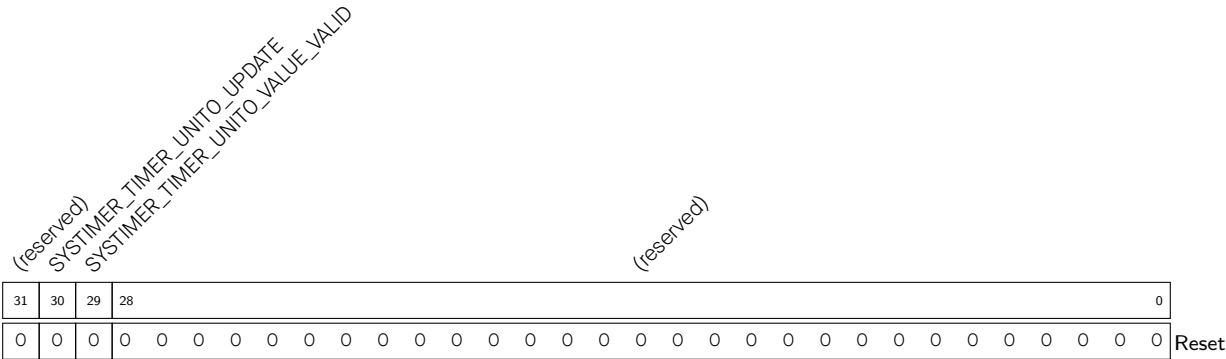
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SYSTIMER_TIMER_UNIT1_WORK_EN Configures whether to enable UNIT1.
0: Disable
1: Enable
(R/W)

SYSTIMER_TIMER_UNIT0_WORK_EN Configures whether to enable UNIT0.
0: Disable
1: Enable
(R/W)

SYSTIMER_CLK_EN Configures register clock gating.
0: Only enable needed clock for register read or write operations.
1: Register clock is always enabled for read and write operations.
(R/W)

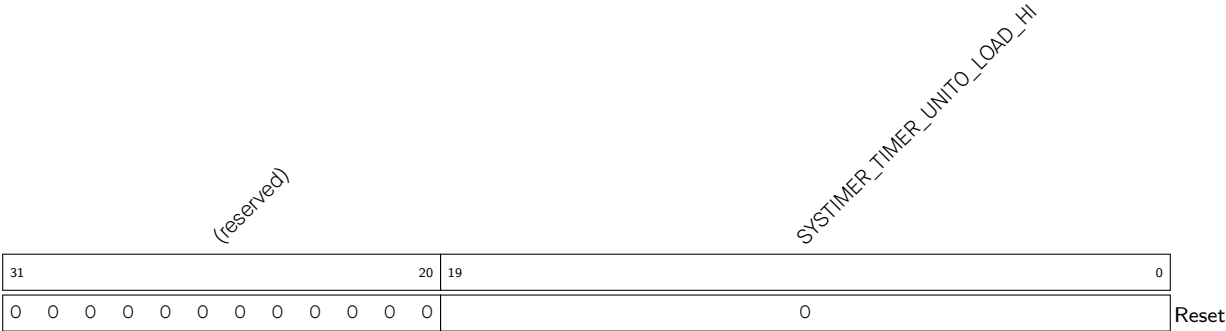
Register 15.2. SYSTIMER_UNIT0_OP_REG (0x0004)



SYSTIMER_TIMER_UNIT0_VALUE_VALID Represents whether UNIT0 value is synchronized and valid.
0: UNIT0 value is neither synchronized nor valid
1: UNIT0 value is synchronized and valid
(R/SS/WT0)

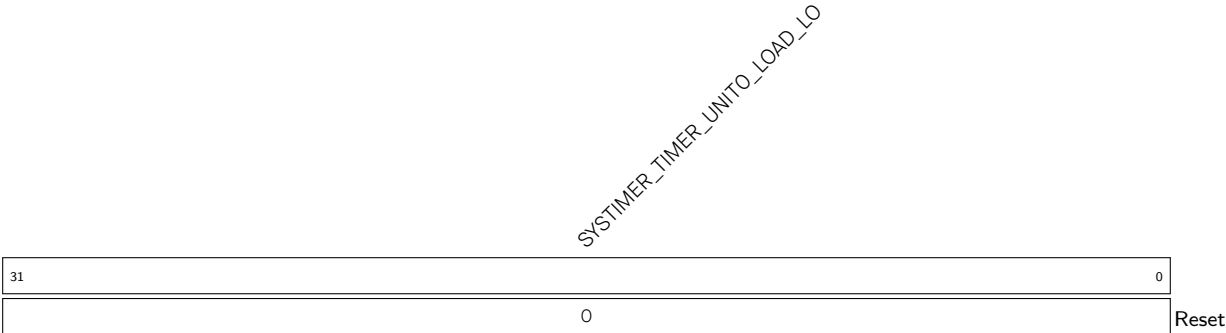
SYSTIMER_TIMER_UNIT0_UPDATE Configures whether to update timer UNIT0, i.e., reads the UNIT0 count value to SYSTIMER_TIMER_UNIT0_VALUE_HI and SYSTIMER_TIMER_UNIT0_VALUE_LO.
0: No effect
1: Update timer UNIT0
(WT)

Register 15.3. SYSTIMER_UNITO_LOAD_HI_REG (0x000C)



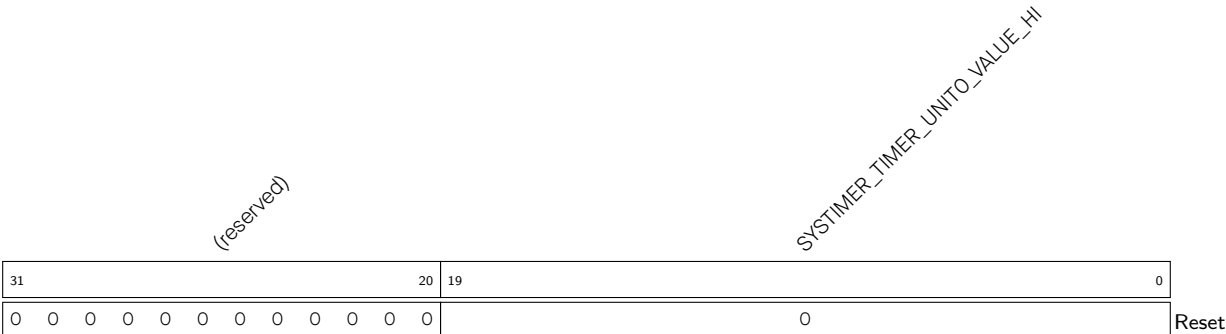
SYSTIMER_TIMER_UNITO_LOAD_HI Configures the value to be loaded to UNITO, high 20 bits. (R/W)

Register 15.4. SYSTIMER_UNITO_LOAD_LO_REG (0x0010)



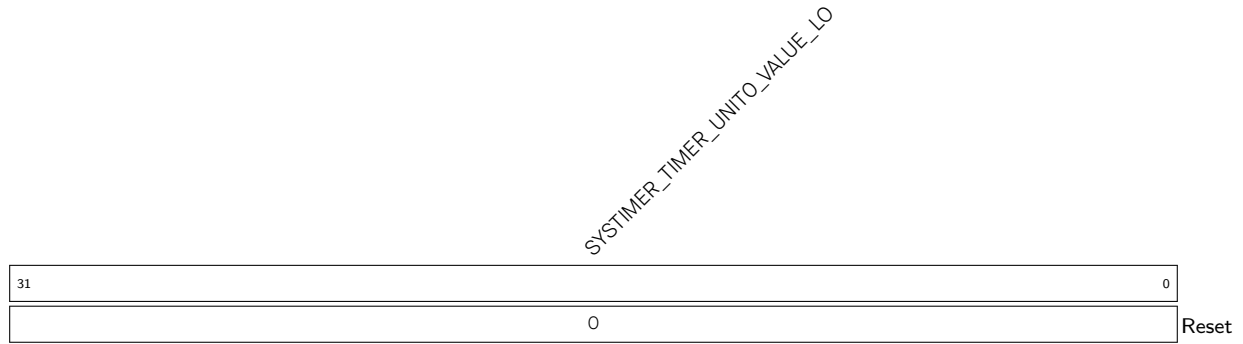
SYSTIMER_TIMER_UNITO_LOAD_LO Configures the value to be loaded to UNITO, low 32 bits. (R/W)

Register 15.5. SYSTIMER_UNITO_VALUE_HI_REG (0x0040)



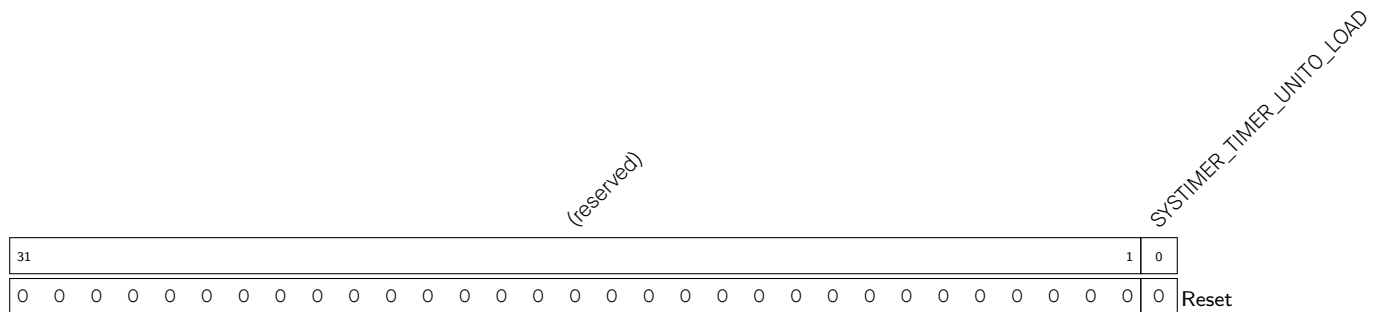
SYSTIMER_TIMER_UNITO_VALUE_HI Represents UNITO read value, high 20 bits. (RO)

Register 15.6. SYSTIMER_UNIT0_VALUE_LO_REG (0x0044)



SYSTIMER_TIMER_UNIT0_VALUE_LO Represents UNIT0 read value, low 32 bits. (RO)

Register 15.7. SYSTIMER_UNIT0_LOAD_REG (0x005C)



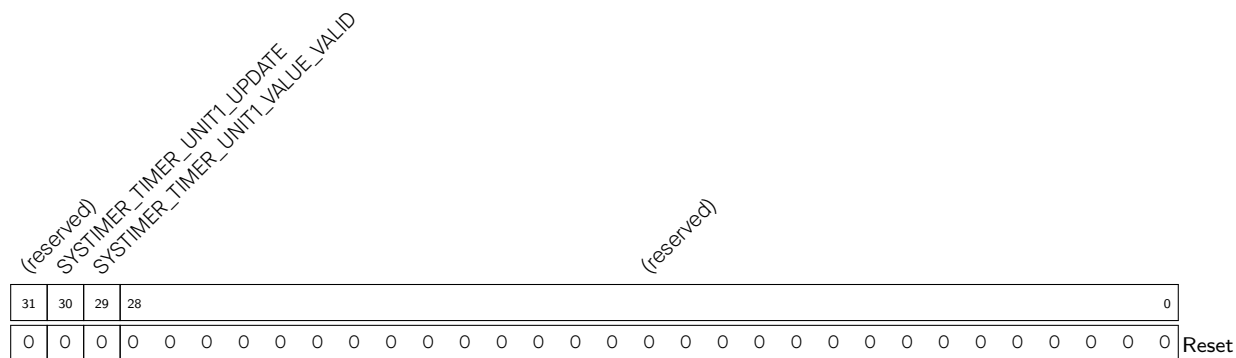
SYSTIMER_TIMER_UNITO_LOAD Configures whether to reload the value of UNITO, i.e., reloads the values of [SYSTIMER_TIMER_UNITO_VALUE_HI](#) and [SYSTIMER_TIMER_UNITO_VALUE_LO](#) to UNITO.

0: No effect

1: Reload the value of UNIT0

(WT)

Register 15.8. SYSTIMER_UNIT1_OP_REG (0x0008)



SYSTIMER_TIMER_UNIT1_VALUE_VALID Represents UNIT1 value is synchronized and valid.
(R/SS/WTC)

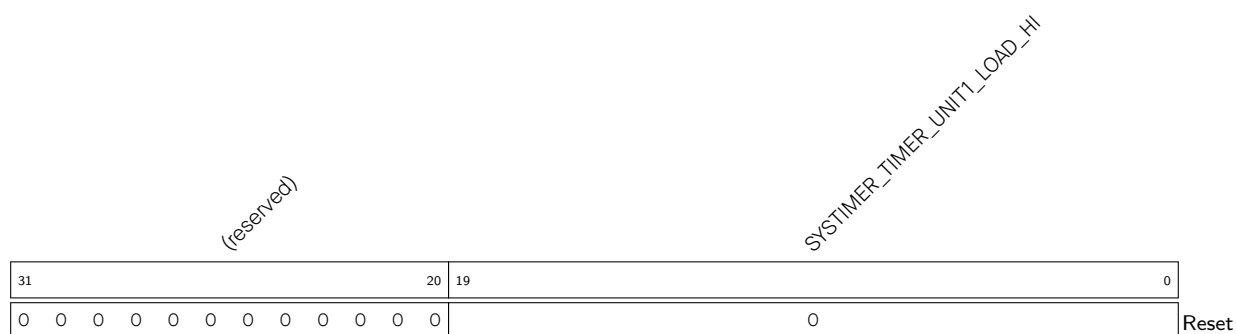
SYSTIMER_TIMER_UNIT1_UPDATE Configures whether to update timer UNIT1, i.e., reads the UNIT1 count value to [SYSTIMER_TIMER_UNIT1_VALUE_HI](#) and [SYSTIMER_TIMER_UNIT1_VALUE_LO](#).

0: No effect

1: Update timer UNIT1

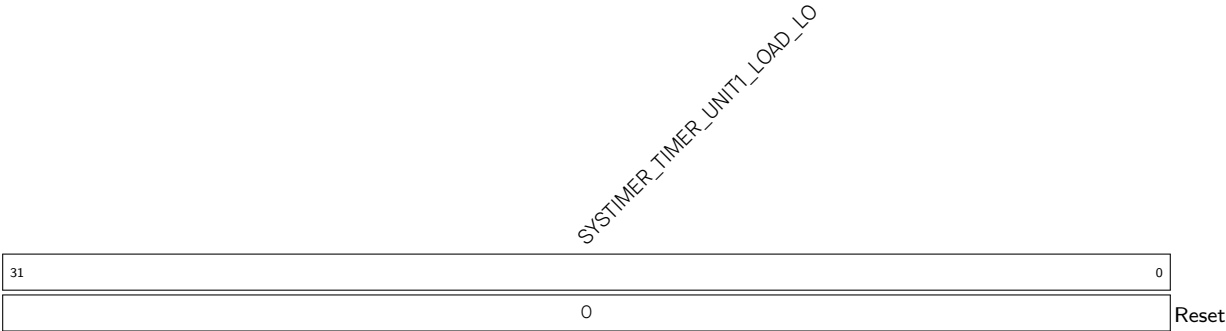
(WT)

Register 15.9. SYSTIMER_UNIT1_LOAD_HI_REG (0x0014)



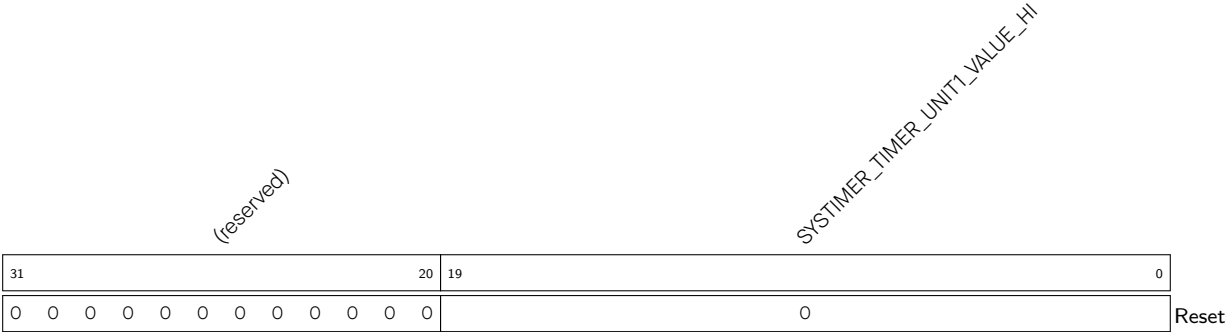
SYSTIMER_TIMER_UNIT1_LOAD_HI Configures the value to be loaded to UNIT1, high 20 bits. (R/W)

Register 15.10. SYSTIMER_UNIT1_LOAD_LO_REG (0x0018)



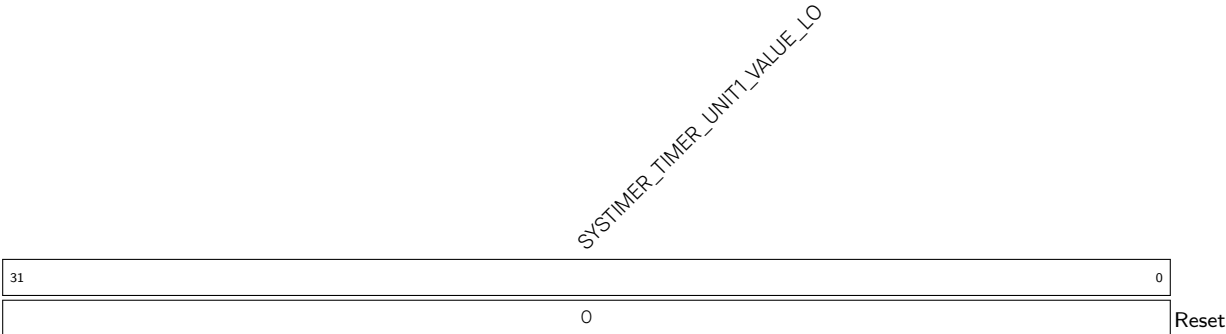
SYSTIMER_TIMER_UNIT1_LOAD_LO Configures the value to be loaded to UNIT1, low 32 bits. (R/W)

Register 15.11. SYSTIMER_UNIT1_VALUE_HI_REG (0x0048)



SYSTIMER_TIMER_UNIT1_VALUE_HI Represents UNIT1 read value, high 20 bits. (RO)

Register 15.12. SYSTIMER_UNIT1_VALUE_LO_REG (0x004C)



SYSTIMER_TIMER_UNIT1_VALUE_LO Represents UNIT1 read value, low 32 bits. (RO)

Register 15.13. SYSTIMER_UNIT1_LOAD_REG (0x0060)

Diagram of the SYSTIMER_TMR_LOAD register structure:

- Bit 31: (reserved)
- Bits 31 to 1: (reserved)
- Bit 0: Reset

SYSTIMER_TIMER_UNIT1_LOAD Configures whether to reload the value of UNIT1, i.e., reload the values of [SYSTIMER_TIMER_UNIT1_VALUE_HI](#) and [SYSTIMER_TIMER_UNIT1_VALUE_LO](#) to UNIT1.

0: No effect

1: Reload the value of UNIT1

(WT)

Register 15.14. SYSTIMER_TARGETO_HI_REG (0x001C)

Diagram illustrating the SYSTIMER_TMR register structure. The register is 32 bits wide, divided into two main sections:

- Bits 31-20:** Labeled "(reserved)".
- Bits 19-0:** Labeled "SYSTIMER_TIMER_TARGET0_HI". This section contains a single '0' at bit 19 and is followed by "Reset".

SYSTIMER_TIMER_TARGETO_HI Configures the alarm value to be loaded to COMP0, high 20 bits.

(R/W)

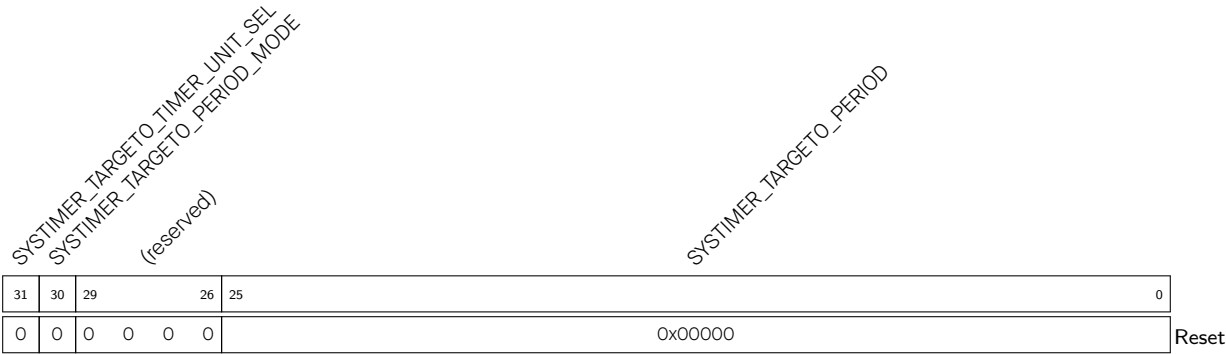
Register 15.15. SYSTIMER_TARGET0_LO_REG (0x0020)

Diagram illustrating the SYSTIMER_TIMCR register structure. The register is 32 bits wide, with bit 31 labeled SYSTIMER_TIMER_TARGET0_LO and bit 0 labeled Reset.

SYSTIMER_TIMER_TARGET0_LO Configures the alarm value to be loaded to COMP0, low 32 bits.

(R/W)

Register 15.16. SYSTIMER_TARGETO_CONF_REG (0x0034)



- SYSTIMER_TARGETO_PERIOD

Configures COMPO alarm period. (R/W)
- SYSTIMER_TARGETO_PERIOD_MODE

Selects the two alarm modes for COMPO.

0: Target mode

1: Period mode

(R/W)
- SYSTIMER_TARGETO_TIMER_UNIT_SEL

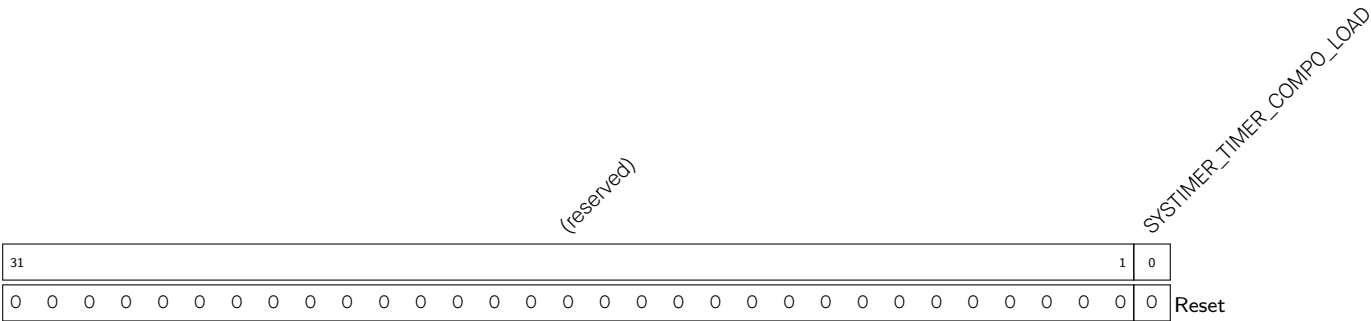
Chooses the count value for comparison with COMPO.

0: Use the count value from UNIT0

1: Use the count value from UNIT1

(R/W)

Register 15.17. SYSTIMER_COMPO_LOAD_REG (0x0050)



- SYSTIMER_TIMER_COMPO_LOAD

Configures whether to enable COMPO synchronization, i.e., reload the alarm value/period to COMPO.

0: No effect

1: Enable COMPO synchronization

(WT)

Register 15.18. SYSTIMER_TARGET1_HI_REG (0x0024)

Diagram of the SYSTIMER_TARGET1 register structure:

- Bits 31 to 20: (reserved)
- Bits 19 to 0: SYSTIMER_TARGET1_HI
- Bit 0: Reset

SYSTIMER_TIMER_TARGET1_HI Configures the alarm value to be loaded to COMP1, high 20 bits.
(R/W)

Register 15.19. SYSTIMER TARGET1 LO REG (0x0028)

Diagram of the SYSTIMER_TARGET1_LO register. The register is 32 bits wide, with bit 31 on the left and bit 0 on the right. Bit 0 is labeled "Reset".

SYSTIMER_TIMER_TARGET1_LO Configures the alarm value to be loaded to COMP1, low 32 bits.
(R/W)

Register 15.20. SYSTIMER_TARGET1_CONF_REG (0x0038)

Diagram illustrating the structure of the SYSTIMER_TARGET1 register:

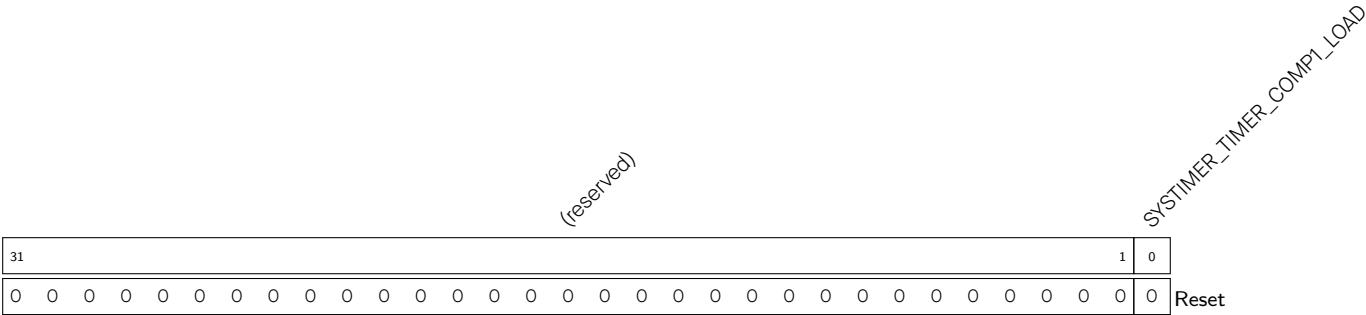
- Bits 31-29: SYSTIMER_TARGET1_TIMER_UNIT_SEL
- Bits 28-26: SYSTIMER_TARGET1_PERIOD_MODE
- Bits 15-0: SYSTIMER_TARGET1_PERIOD

SYSTIMER_TARGET1_PERIOD Configures COMP1 alarm period. (R/W)

SYSTIMER_TARGET1_PERIOD_MODE Selects the two alarm modes for COMP1. See details in [SYSTIMER_TARGET0_PERIOD_MODE](#). (R/W)

SYSTIMER_TARGET1_TIMER_UNIT_SEL Chooses the count value for comparison with COMP1. See details in [SYSTIMER_TARGET0_TIMER_UNIT_SEL](#). (R/W)

Register 15.21. SYSTIMER_COMP1_LOAD_REG (0x0054)

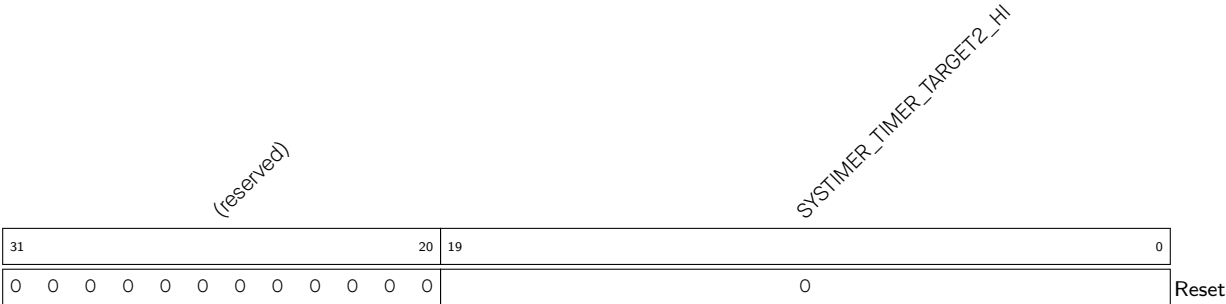


SYSTIMER_TIMER_COMP1_LOAD Configures whether to enable COMP1 synchronization, i.e., reload the alarm value/period to COMP1.

0: No effect

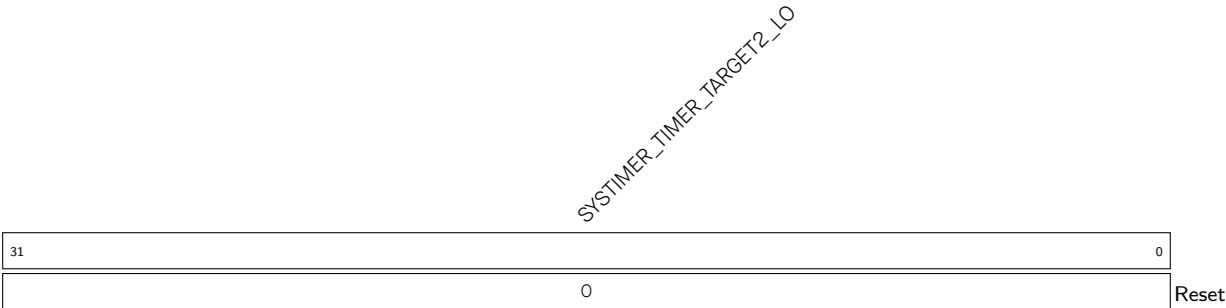
1: Enable COMP1 synchronization (WT)

Register 15.22. SYSTIMER_TARGET2_HI_REG (0x002C)



SYSTIMER_TIMER_TARGET2_HI Configures the alarm value to be loaded to COMP2, high 20 bits. (R/W)

Register 15.23. SYSTIMER_TARGET2_LO_REG (0x0030)



SYSTIMER_TIMER_TARGET2_LO Configures the alarm value to be loaded to COMP2, low 32 bits. (R/W)

Register 15.24. SYSTIMER_TARGET2_CONF_REG (0x003C)

31		30		29		26		25		0													
0		0		0		0		0		0x00000													

Reset

SYSTIMER_TARGET2_PERIOD Configures COMP2 alarm period. (R/W)

SYSTIMER_TARGET2_PERIOD_MODE Configures the two alarm modes for COMP2. See details in [SYSTIMER_TARGET2_PERIOD_MODE](#). (R/W)

SYSTIMER_TARGET2_TIMER_UNIT_SEL Chooses the count value for comparison with COMP2. See details in [SYSTIMER_TARGET2_TIMER_UNIT_SEL](#). (R/W)

Register 15.25. SYSTIMER_COMP2_LOAD_REG (0x0058)

(reserved)																														1		0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Reset

SYSTIMER_TIMER_COMP2_LOAD Configures whether to enable COMP2 synchronization, i.e., reload the alarm value/period to COMP2.

0: No effect

1: Enable COMP2 synchronization

(WT)

Register 15.26. SYSTIMER_INT_ENA_REG (0x0064)

31																													3	2	1	0
(reserved) SYSTEMER_TARGET2_INT_ENA SYSTEMER_TARGET1_INT_ENA SYSTEMER_TARGET0_INT_ENA																																
0 0 0 0 0																																
Reset																																

SYSTIMER_TARGET0_INT_ENA Write 1 to enable SYSTIMER_TARGET0_INT. (R/W)

SYSTIMER_TARGET1_INT_ENA Write 1 to enable SYSTIMER_TARGET1_INT. (R/W)

SYSTIMER_TARGET2_INT_ENA Write 1 to enable SYSTIMER_TARGET2_INT. (R/W)

Register 15.27. SYSTIMER_INT_RAW_REG (0x0068)

[illegible]

SYSTIMER_TARGETO_INT_RAW The raw interrupt status of SYSTIMER_TARGETO_INT. (R/WTC/SS)

SYSTIMER_TARGET1_INT_RAW The raw interrupt status of SYSTIMER_TARGET1_INT. (R/WTC/SS)

SYSTIMER_TARGET2_INT_RAW The raw interrupt status of SYSTIMER_TARGET2_INT. (R/WTC/SS)

Register 15.28. SYSTIMER_INT_CLR_REG (0x006C)

Diagram of the 32-bit SYSTEMER register structure:

- Bit 31: (reserved)
- Bits 28-30: SYSTEMER_TARGET2_INT_CLR (3 bits)
- Bits 26-27: SYSTEMER_TARGET1_INT_CLR (2 bits)
- Bit 25: SYSTEMER_TARGET0_INT_CLR (1 bit)

Reset signal is applied to the register, setting all bits to 0.

SYSTIMER_TARGET0_INT_CLR Write 1 to clear SYSTIMER_TARGET0_INT. (WT)

SYSTIMER_TARGET1_INT_CLR Write 1 to clear SYSTIMER_TARGET1_INT. (WT)

SYSTIMER_TARGET2_INT_CLR Write 1 to clear SYSTIMER_TARGET2_INT. (WT)

Register 15.29. SYSTIMER_INT_ST_REG (0x0070)

Diagram illustrating the structure of the 32-bit Reset signal. The signal is divided into four 8-bit fields. The first 24 bits are labeled "(reserved)". The last 8 bits are labeled "Reset" and are further divided into four 2-bit fields: SYSTEMER_TARGET2_INT_ST, SYSTEMER_TARGET1_INT_ST, SYSTEMER_TARGET0_INT_ST, and SYSTEMER_TARGET0_INT_ST.

SYSTIMER_TARGETO_INT_ST The masked interrupt status of SYSTIMER_TARGETO_INT. (RO)

SYSTIMER_TARGET1_INT_ST The masked interrupt status of SYSTIMER_TARGET1_INT. (RO)

SYSTIMER_TARGET2_INT_ST The masked interrupt status of SYSTIMER_TARGET2_INT. (RO)

Register 15.30. SYSTIMER_REAL_TARGET0_LO_REG (0x0074)

Diagram of the SYSTEM_TARGET0_LO_RO register. The register is 32 bits wide, with bit 31 on the left and bit 0 on the right. Bit 0 is labeled "Reset". The label "SYSTEM_TARGET0_LO_RO" is written diagonally across the register.

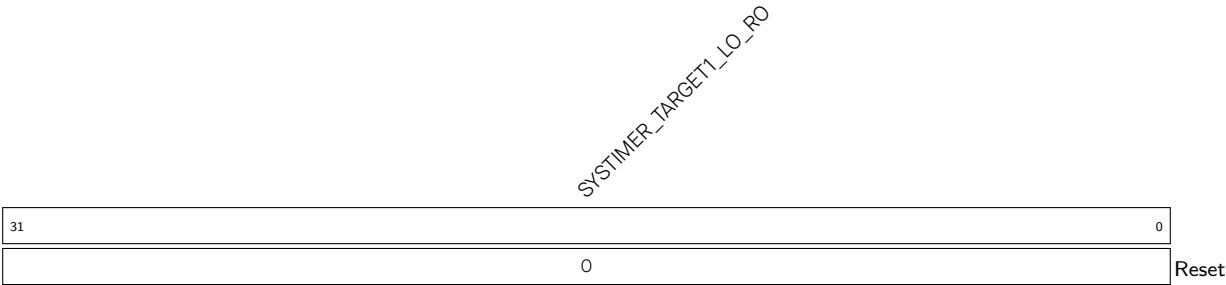
SYSTIMER_TARGET0_LO_RO Represents the actual target value of COMP0, low 32 bits. (RO)

Register 15.31. SYSTIMER_REAL_TARGET0_HI_REG (0x0078)



SYSTIMER_TARGET0_HI_RO Represents the actual target value of COMP0, high 20 bits. (RO)

Register 15.32. SYSTIMER_REAL_TARGET1_LO_REG (0x007C)



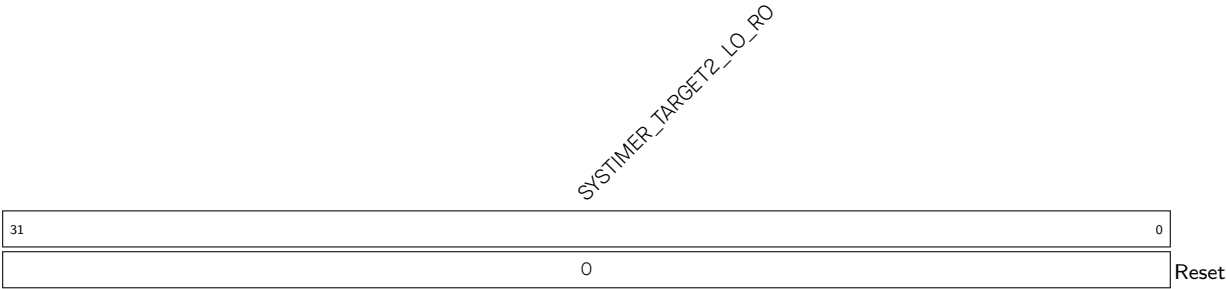
SYSTIMER_TARGET1_LO_RO Represents the actual target value of COMP1, low 32 bits. (RO)

Register 15.33. SYSTIMER_REAL_TARGET1_HI_REG (0x0080)



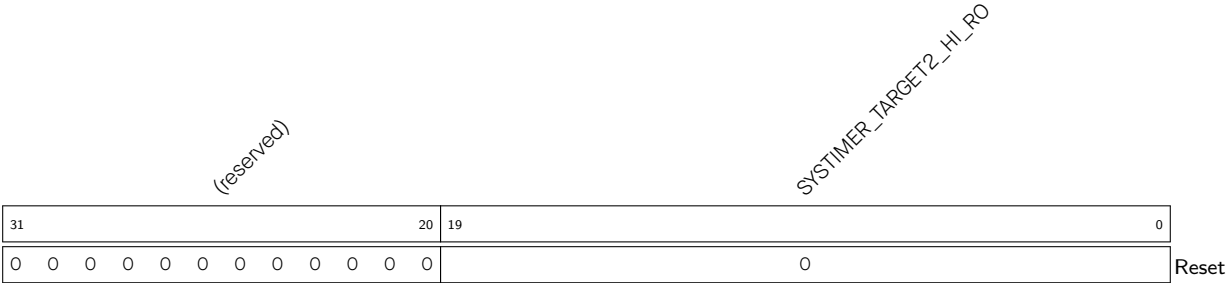
SYSTIMER_TARGET1_HI_RO Represents the actual target value of COMP1, high 20 bits. (RO)

Register 15.34. SYSTIMER_REAL_TARGET2_LO_REG (0x0084)



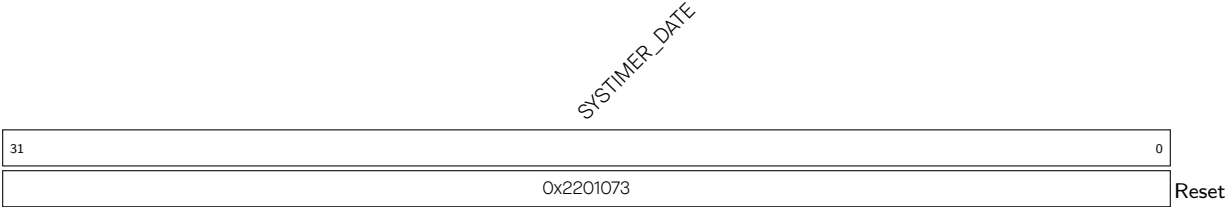
SYSTIMER_TARGET2_LO_RO Represents the actual target value of COMP2, low 32 bits. (RO)

Register 15.35. SYSTIMER_REAL_TARGET2_HI_REG (0x0088)



SYSTIMER_TARGET2_HI_RO Represents the actual target value of COMP2, high 20 bits. (RO)

Register 15.36. SYSTIMER_DATE_REG (0x00FC)



SYSTIMER_DATE Version control register. (R/W)